



VISUAL ANALYTICS: SYSTEMS FOR MODELING

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TODAY'S PLAN

- Modeling overview
- VA systems for model:
 - Explanation
 - Debugging
 - Construction and steering
 - Comparison
- Exploratory model analysis (EMA)
 - Definition
 - Contrast with EDA
 - Workflow

MODELING OVERVIEW: WARM UP

- Give an example of a predictive model
- What is the goal of a predictive model?
- How are predictive models generated?
- How are predictive models evaluated?

MODELING OVERVIEW

- In machine learning, a **model** is what is being learned from the data to solve a specific task
- Tasks include:
 - Classification
 - Regression
 - Clustering
 - Association discovery

MODELING OVERVIEW

Three general types of models

1. Geometric

- Use geometric concepts (ex. lines, planes, distance) to classify input
- Ex. Basic linear classifiers, Nearest-neighbor classifiers

2. Probabilistic

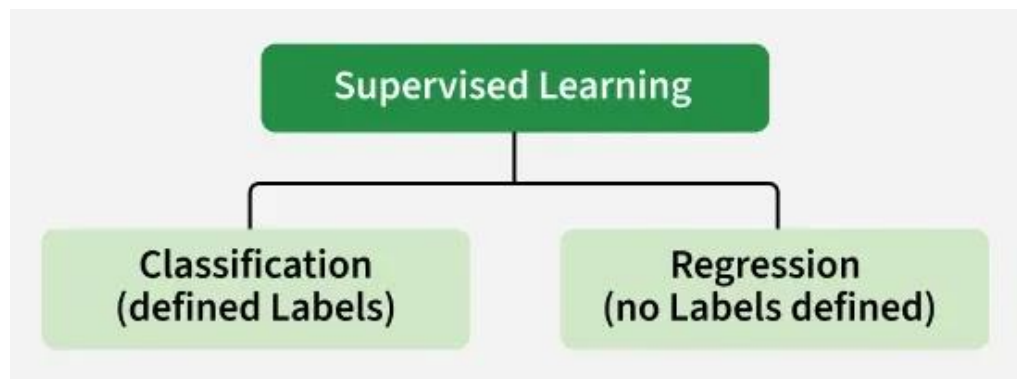
- Attempts to uncover underlying probability distributions that determine how variables are related
- Ex. Bayes classifier

3. Logical

- Algorithmic in nature; a series of if-then statements
- Ex. Decision tree

MODELING OVERVIEW

- We will focus on supervised machine learning

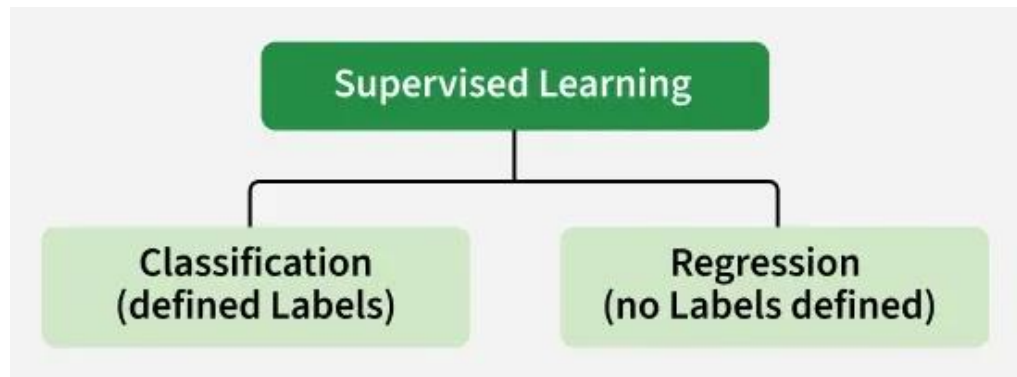


User ID	Gender	Age	Salary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	1
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	1
15728773	Male	27	58000	1
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	1
15727311	Female	35	65000	0
15570769	Female	26	80000	1
15606274	Female	26	52000	0
15746139	Male	20	86000	1
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1

Figure A: CLASSIFICATION

MODELING OVERVIEW

- We will focus on supervised machine learning



Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
10.69261758	986.882019	54.19337313	195.7150879	3.278597116
13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
17.76400389	985.4262085	33.54900114	280.7827454	2.343950987
11.25680746	988.9386597	53.74139903	68.15406036	1.650191426
14.37810685	989.6819458	40.70884681	72.62069702	1.553469896
18.45114201	990.2960205	30.85038484	71.70604706	1.005017161
22.54895853	989.9562988	22.81738811	44.66042709	0.264133632
24.23155922	988.796875	19.74790765	318.3214111	0.329656571

Figure B: REGRESSION

MODELING OVERVIEW

- We will focus on supervised machine learning
- Process:
 1. Collect Labeled Data
 2. Split the Dataset
 3. Train the Model
 4. Validate and Test the Model
 5. Deploy and Predict on New Data

MODELING OVERVIEW

- We will focus on supervised machine learning
- Use cases:
 - Fraud Detection in Banking
 - Parkinson Disease Prediction
 - Customer Churn Prediction
 - Cancer cell classification
 - Stock Price Prediction

Where is modeling used for decision making?

MODELING OVERVIEW

- Evaluation of classification models

- Confusion matrix:

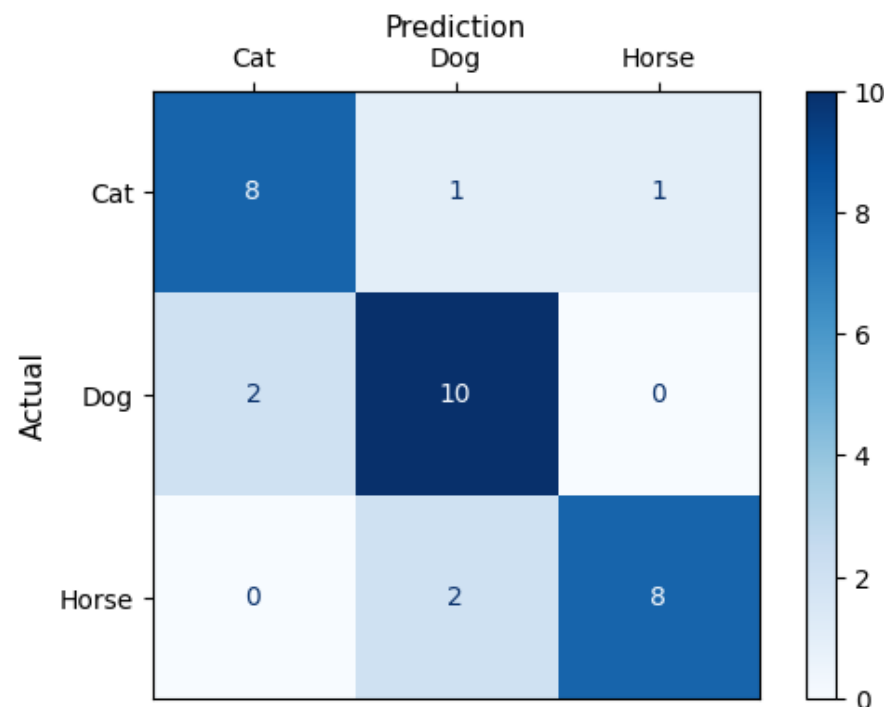
	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

- Accuracy $\frac{TP+TN}{TP+TN+FP+FN}$

- Precision $\frac{TP}{TP+FP}$

- Recall $\frac{TP}{TP+FN}$

Confusion Matrix



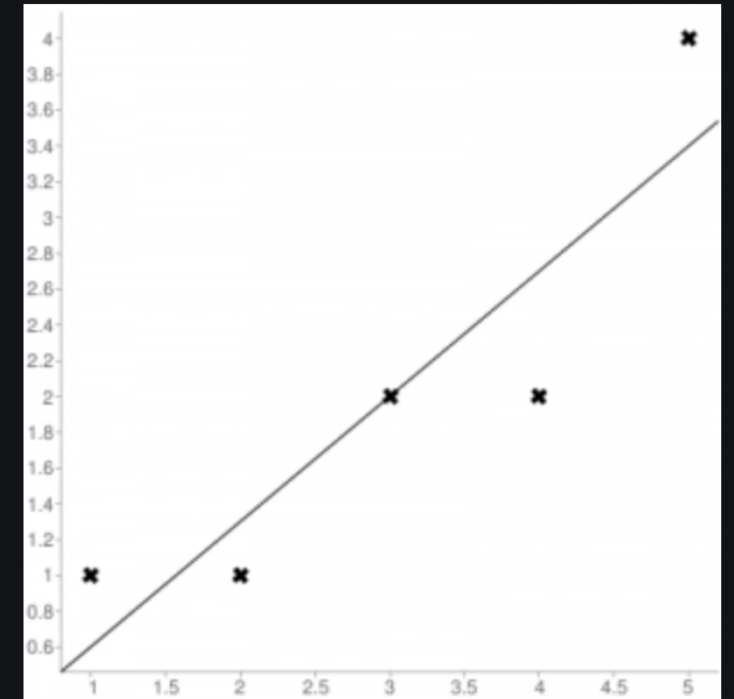
MODELING OVERVIEW

- Evaluation of regression models

- Mean Squared Error: $\frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2$

- R-Squared: $1 - \frac{\sum_{j=1}^n (y_j - \hat{y}_j)^2}{\sum_{j=1}^n (y_j - \bar{y})^2}$

Example: Consider the given data points: (1,1), (2,1), (3,2), (4,2), (5,4).



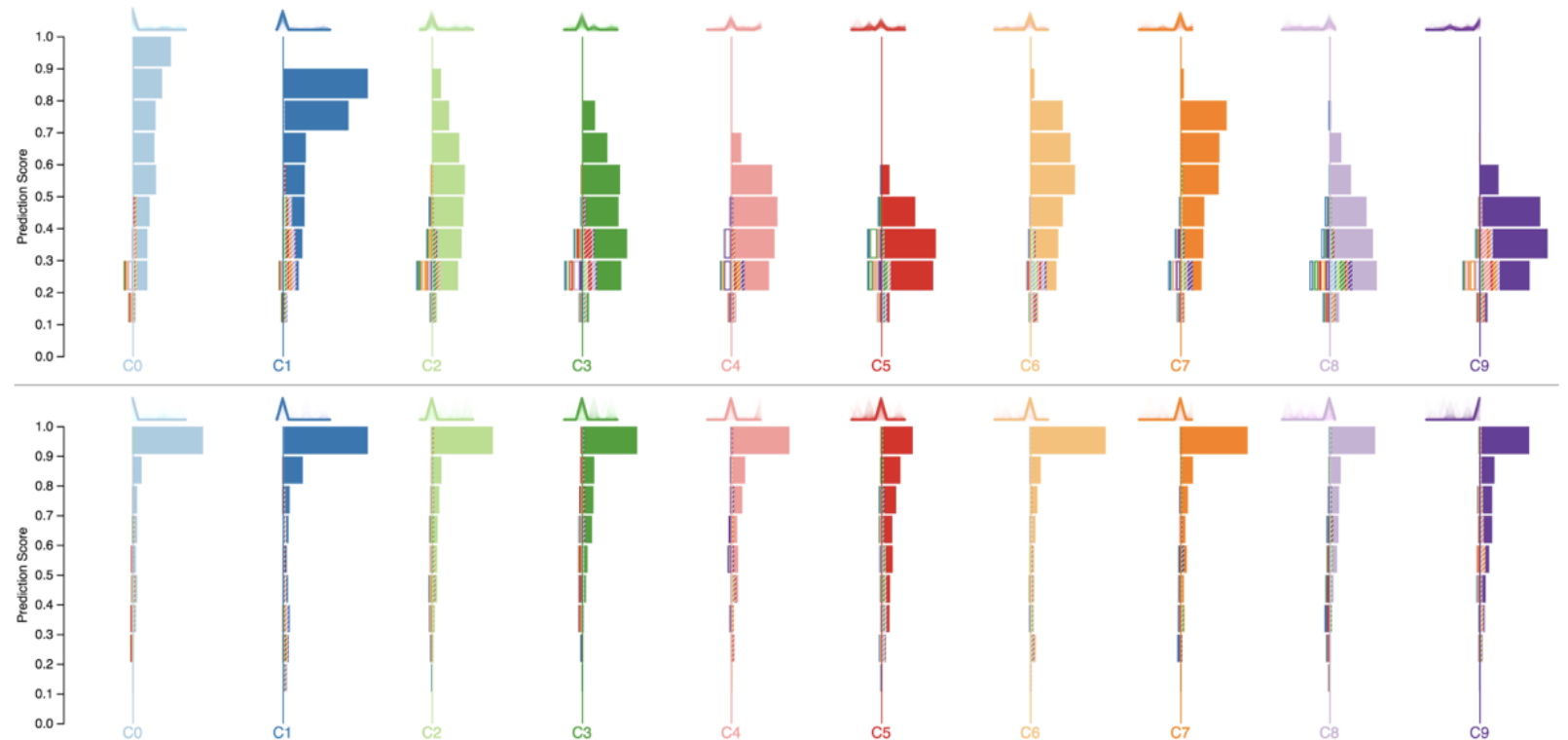
Regression line equation: $Y = 0.7X - 0.1$

SYSTEMS FOR MODEL EXPLANATION

- The explainability of a model is not only important to the model builder themselves, but to anyone else affected by that model, as required by ethical and legal guidelines such as the European Union's General Data Protection Regulation (GDPR)
- Approaches:
 - Systems have predictions by **How might you explain a model's decision to someone without analytic expertise?** learning model makes predicts poorly
 - Simplified classes of models could be used in a VA application to trade off some performance in exchange for a more explainable analysis
 - Instance-level explanations of output

SYSTEMS FOR MODEL EXPLANATION

- Ex. Squares
- <https://www.microsoft.com/en-us/research/video/squares-supporting-interactive-performance-analysis-multiclass-classifiers-2/>



D. Ren, et al.. "Squares: Supporting Interactive Performance Analysis for Multiclass Classifiers," 2017, doi: 10.1109/TVCG.2016.2598828

SYSTEMS FOR MODEL DEBUGGING

- While the calculations used in machine learning models can be excessively complicated, endemic properties of models that cause poor predictions can sometimes be diagnosed visually relatively easily
- These tools allow a model builder to view *how* and *where* their model is breaking, on specified data instances

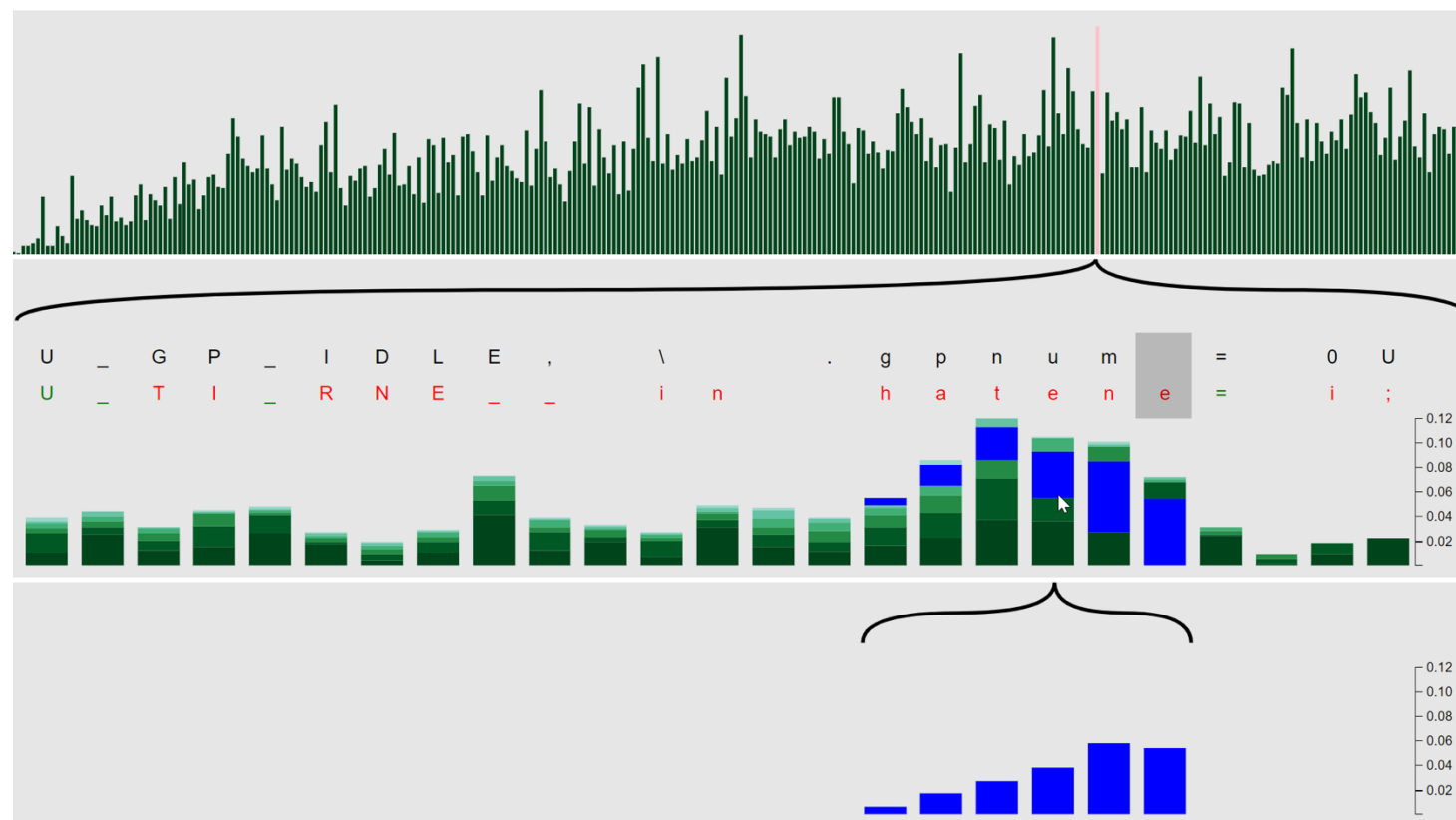
SYSTEMS FOR MODEL DEBUGGING

RNNbow

Visualizing Learning via Backpropagation Gradients in Recurrent Neural Networks

Batch Number: 222
Training Cells: 555000 - 555024
Maximum Gradient: 0.1182

■ Ex. RNNbow



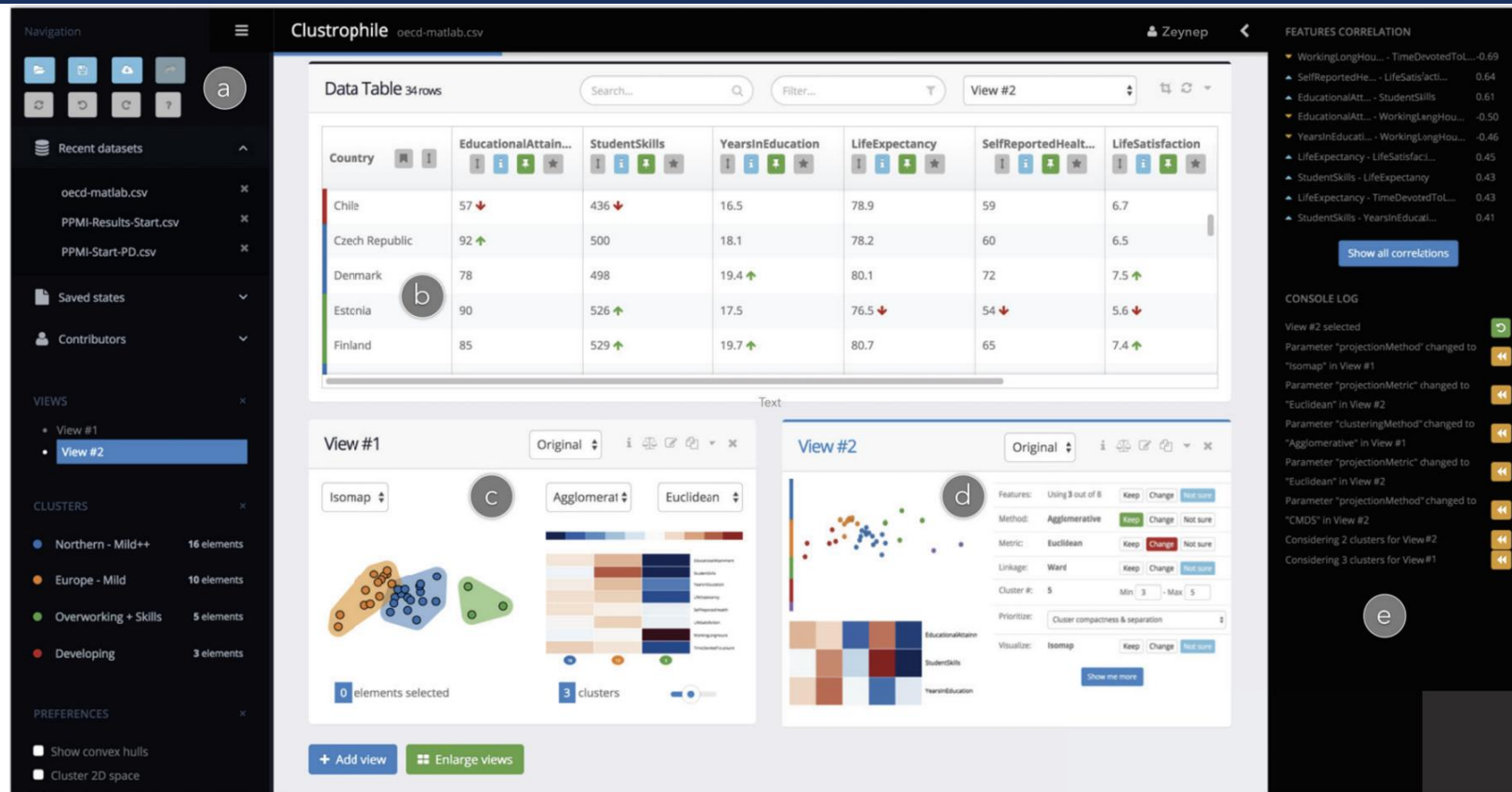
Cashman et al.. (2019), RNNbow: Visualizing Learning Via Backpropagation Gradients in RNNs.

SYSTEMS FOR MODEL CONSTRUCTION AND STEERING

- When building a model, a modeling expert modifies various hyperparameters to maximize some utility function, whether explicitly or implicitly defined
- Visual analytics systems can assist domain experts to control the model fitting process by allowing the user to directly manipulate the model's hyperparameters or by inferring the model's hyperparameters through observing and analyzing the user's interactions with the visualization

SYSTEMS FOR MODEL CONSTRUCTION AND STEERING

- Ex. Clustrophile
- <https://www.youtube.com/watch?v=UJzgrGIKMMI>



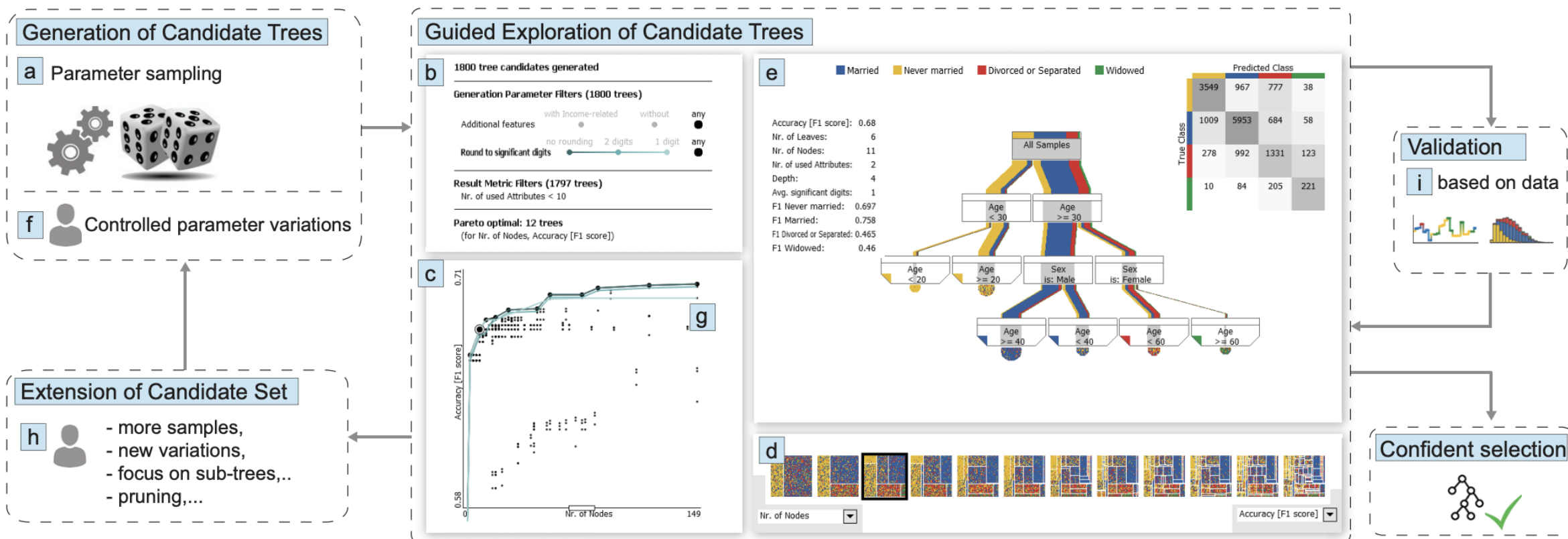
Cavallo and Cagatay.. (2019), Clustrophile 2: Guided Visual Clustering Analysis. doi: 10.1109/TVCG.2018.2864477

SYSTEMS FOR MODEL COMPARISON

- The choice of which model to use from a set of candidate models is highly dependent on the needs of the user and the deployment scenario of a model
- For example, when might we want to minimize false positives vs false negatives?

SYSTEMS FOR MODEL COMPARISON

■ Ex. TreePOD



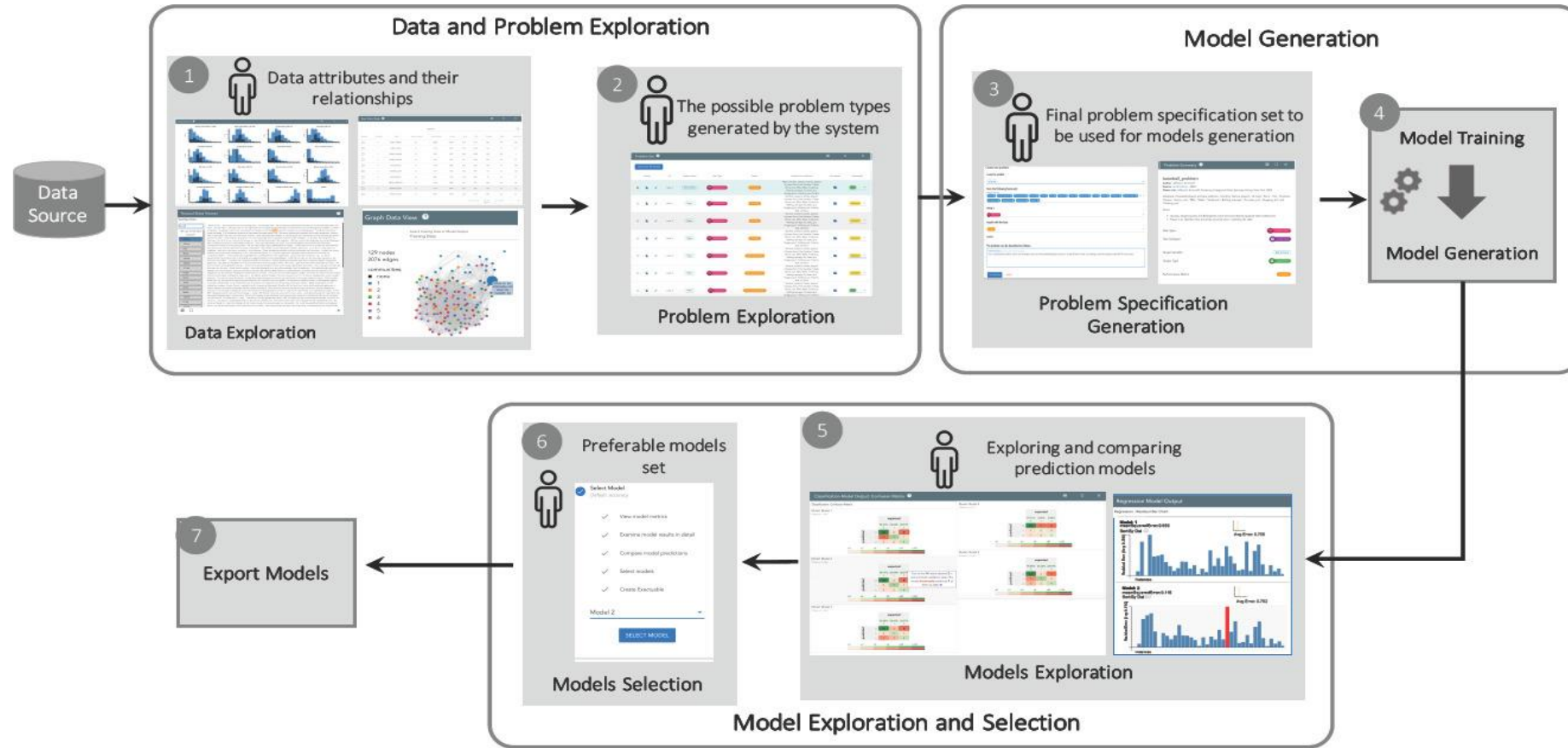
EMA: DEFINITION

- In the modern era of big data, machine learning, and AI, visual analytics systems have begun to take on a new role: to help the user in refining machine learning models
- Exploratory Model Analysis (EMA) is the process of exploring the set of potential models that can be trained on a given set of data
- The goal of EMA is twofold:
 - Discover variables in the dataset on which reliable predictions can be made
 - Find the most suitable and robust types of models to predict these variables

EMA VS. EDA

- Broker performing EDA
 - Goal:
 - Understand current patterns in financial data; i.e. current market landscape
 - Outcomes:
 - Decision
 - Data item (ex. an anomaly)
 - Interesting pattern in the data
 - Data is the focus
- Quantitative analyst performing EMA
 - Goal:
 - Understand what types of predictions are possible given the collected financial data
 - Outcomes:
 - Model (or set of models)
 - Hyperparameters
 - Properties of predictions on hold out data
 - Model is the focus

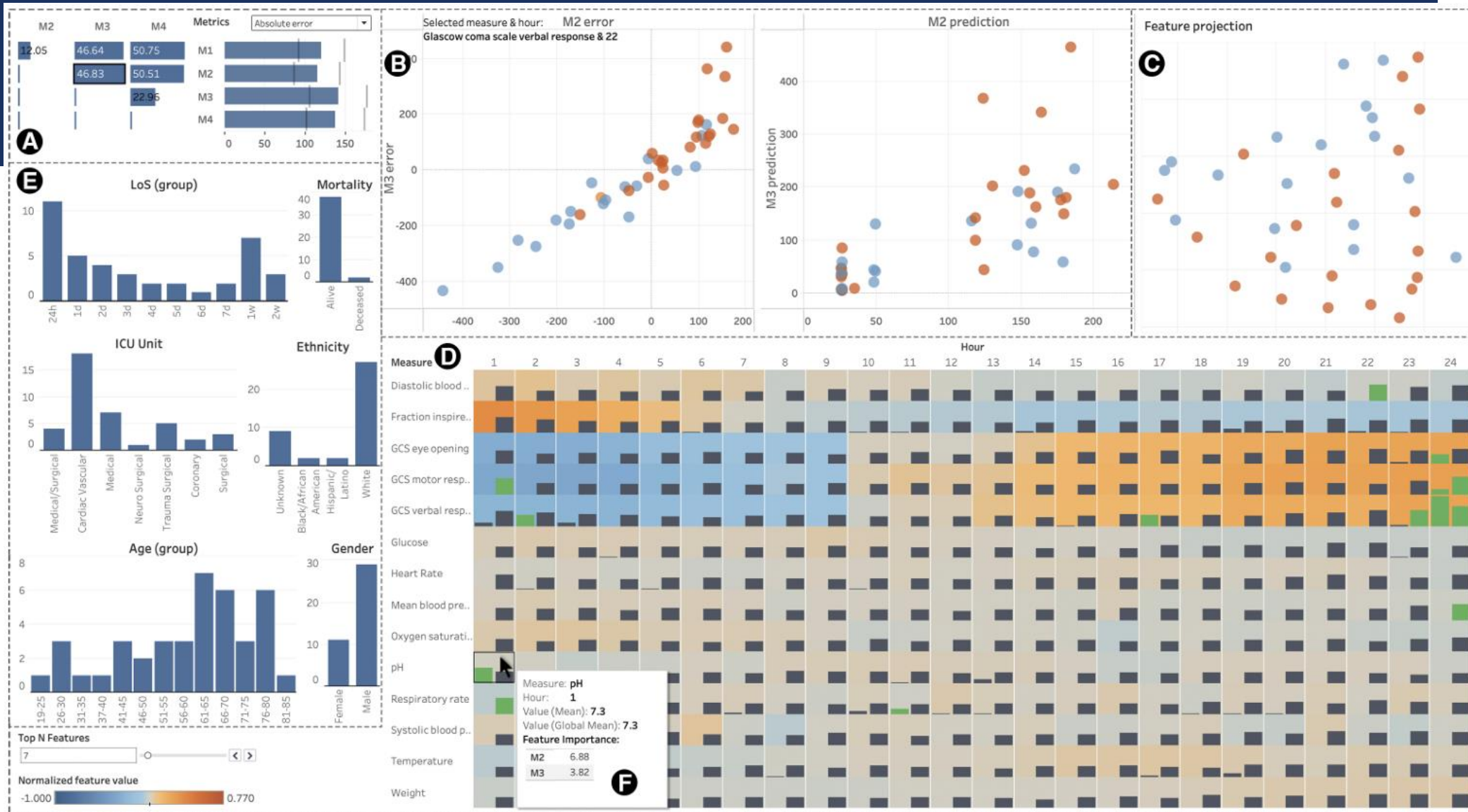
EMA WORKFLOW



Cashman et al.. (2019), A User-based Visual Analytics Workflow for Exploratory Model Analysis. <https://doi.org/10.1111/cgf.13681>

EMA: EX.

VMS



YOUR TURN!

- Choose a system from this visual survey of VA+ML systems: <https://trustmlvis.lnu.se/>
- Dig more deeply into what the system visualizes
 - What model technique(s) is it built for?
 - What does it visualize and how?
 - Would you make any changes to the design? Why or why not?